and, for a vector function $\boldsymbol{w} : \mathcal{D} \to \mathbb{R}^{\mathrm{N_c}}$, we have

$$\|\boldsymbol{w}\|^2_{L^2(\mu)} = \sum_{i=1}^{\mathrm{N_c}} \|w_i\|^2_{L^2(\mu)}.$$

With this in mind, we state a useful property that we will need in the next chapter in order to clarify results on the error in invariant measures.

Proposition 6.4 *Let U be C^∞ on the position domain $\mathcal{Q} = \mathcal{T}^{\mathrm{N_c}}$ and suppose $g \in \mathcal{H}^1(\mu)$ satisfies*

$$\int_{\mathcal{D}} g \mathrm{d}\mu_\beta = 0,$$

where μ_β is the canonical measure, then the Poisson equation

$$\mathcal{L}^\dagger_{\mathrm{LD}} \varphi = g$$

has a solution $\varphi \in \mathcal{H}^1(\mu)$ which is unique up to addition of a constant multiple of the Gibbs density.

The condition is a consequence of the Fredholm Alternative which says that, for a linear operator $\mathcal{A}$ with compact resolvent, we have one of the following two conditions:

- $\mathcal{A}\varphi = g$ has a solution.
- There is a function y with $\mathcal{A}^\dagger y = 0$ and $\langle y, g \rangle \neq 0$.

In our case, $\mathcal{A} = \mathcal{L}^\dagger_{\mathrm{LD}}$. The only solutions of $\mathcal{L}_{\mathrm{LD}} y = 0$ (conserved quantities of Langevin dynamics) are the constants. So the condition of the proposition eliminates the second alternative, leaving only the first.

If we have two solutions φ_1, φ_2 of $\mathcal{L}^\dagger_{\mathrm{LD}} \varphi = g$, then their difference w must be a solution of $\mathcal{L}^\dagger_{\mathrm{LD}} w = 0$ and the only such solution is a constant multiple of the Gibbs density ρ_β.

More properties of the operator $\mathcal{L}_{\mathrm{LD}}$ are collected in [223].

6.4.5 Convergence of Averages

Given a collection of phase space points distributed according to the density $\rho_0(\boldsymbol{q}, \boldsymbol{p})$, we know that its evolution (under Langevin dynamics) after a time t will be described by

$$\rho(\boldsymbol{q}, \boldsymbol{p}, t) = e^{t\mathcal{L}^\dagger_{\mathrm{LD}}} \rho_0(\boldsymbol{q}, \boldsymbol{p}).$$

More precisely this relation should be interpreted in the variational sense, so that we have, for any suitable test function ϕ, e.g. $\phi \in C^\infty$,

$$\int_{\mathcal{D}} \phi \rho(\boldsymbol{q},\boldsymbol{p},t)\mathrm{d}\omega = \int_{\mathcal{D}} \phi e^{t\mathcal{L}_{\mathrm{LD}}^\dagger}\rho_0(\boldsymbol{q},\boldsymbol{p})\mathrm{d}\omega.$$

This is of course an inner product, and due to the properties of the inner product, we may interpret the right hand side as

$$\int_{\mathcal{D}} \left[e^{t\mathcal{L}_{\mathrm{LD}}}\phi(\boldsymbol{q},\boldsymbol{p})\right]\rho_0(\boldsymbol{q},\boldsymbol{p})\mathrm{d}\omega,$$

which says that the average of a function ϕ with respect to the distribution at time t is equivalent to the average of the forward propagation of ϕ with respect to the initial distribution.

How rapidly an average converges will depend on the spectral properties of the operator as well as the choice of the initial distribution. For example, if we assume that $\lambda_1, \rho^{(1)}$ is some eigenvalue, eigenfunction pair with $\mathrm{Re}(\lambda_1) < 0$ and take the initial distribution to be

$$\rho_0 = \rho_{\mathrm{eq}} + \alpha\rho^{(1)},$$

for $\alpha \in \mathbb{R}$, then we have

$$\begin{aligned}\rho(\boldsymbol{q},\boldsymbol{p},t) = e^{t\mathcal{L}_{\mathrm{LD}}^\dagger}\rho_0 &= e^{t\mathcal{L}_{\mathrm{LD}}^\dagger}\rho_{\mathrm{eq}} + \alpha e^{t\mathcal{L}_{\mathrm{LD}}^\dagger}\rho^{(1)} \\ &= \rho_{\mathrm{eq}} + \alpha e^{t\lambda_1}\rho^{(1)}.\end{aligned}$$

Since all the eigenvalues lie to the left of a vertical line in the left half plane, for any ϕ in C^∞ we have

$$\int_{\mathcal{D}} \phi\rho(\boldsymbol{q},\boldsymbol{p},t)\mathrm{d}\omega = \mathrm{Av}_\beta(\phi) + \alpha e^{t\lambda_1}\int_{\mathcal{D}} \phi\rho^{(1)}\mathrm{d}\omega.$$

The latter integral is bounded since both ϕ and $\rho^{(1)}$ are in L^2, and since $\mathrm{Re}(\lambda_1) < 0$, we see that this converges exponentially to the desired average.